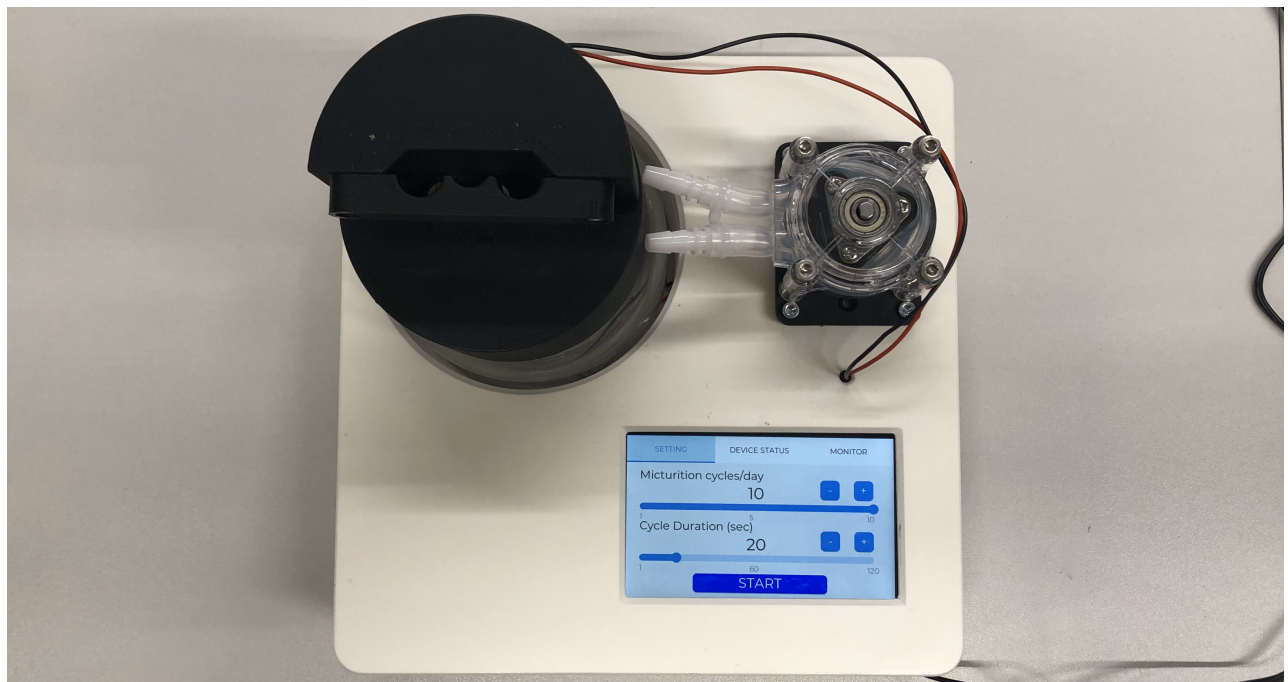


User Interface and Control Software for a Automated Test Bench for the Analysis of Biofilm Formation

Project Thesis in Computational Engineering

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Declaration

I hereby declare that I have written this thesis without outside help and without using any sources other than those indicated. This thesis has not been submitted in the same or a similar version to any other examination office and been accepted as part of an examination performance. All statements that have been adopted literally or paraphrased are marked as such.

Erlangen, 30.11.2025

I-En Lee

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1 Introduction and objective

Urinary incontinence is a common urological condition that significantly affects quality of life. According to the World Health Organization and large-scale epidemiological studies, hundreds of millions of people worldwide are affected. In Germany, surveys show that 8–10% of the adult population experiences Urinary incontinence, with stress urinary incontinence (SUI) being the most common type. It is estimated that around 3 million people in Germany suffer from SUI, and women are more than three times as likely to be affected as men. This condition not only affects daily life and social interactions but also causes psychological stress and increases healthcare costs.[1]

For moderate to severe cases, medical devices are often required to help maintain continence. However, current artificial urinary sphincters (AUS), while considered a standard treatment option, have notable limitations such as high revision rates, the need for invasive surgery, non-intuitive operation, and poor suitability for female anatomy [2]. These limitations highlight the urgent need for improved device designs and dedicated testing platforms.

1.1 Overview of Stress Urinary Incontinence (SUI)

SUI occurs when bladder pressure exceeds urethral pressure during activities that increase abdominal pressure (e.g., coughing or physical exertion), and is caused by factors such as dysfunction of the detrusor muscle, pelvic floor support, or neural control of storage and voiding. Treatment options range from lifestyle modifications, pelvic floor muscle training, and medications to surgical procedures supporting the urethra or enhancing bladder capacity [3]. However, although AUS implantation may provide durable continence, it involves surgical risks, steep learning curve for use, and a high likelihood of needing revision surgery within several years post-implantation [4].

1.2 Limitations of Current AUS

As shown in Figure 1, a traditional AUS consists of a cuff around the urethra, a control pump, and a pressure-regulating balloon. The patient must manually squeeze the pump to urinate. While clinically effective, multiple limitations have been reported in the literature, including high revision rates, the need for invasive surgery, non-intuitive operation, and limited female compatibility [6]:

1. High revision rates: Long-term studies report that a considerable proportion of devices require replacement within 10 years due to mechanical failure, infection, or erosion.
2. Invasive surgery: Requires general or spinal anesthesia and hospitalization, with extended recovery times.
3. Non-intuitive operation: Difficult for elderly patients or those with reduced hand dexterity.
4. Limited female compatibility: Most designs are based on male anatomy.

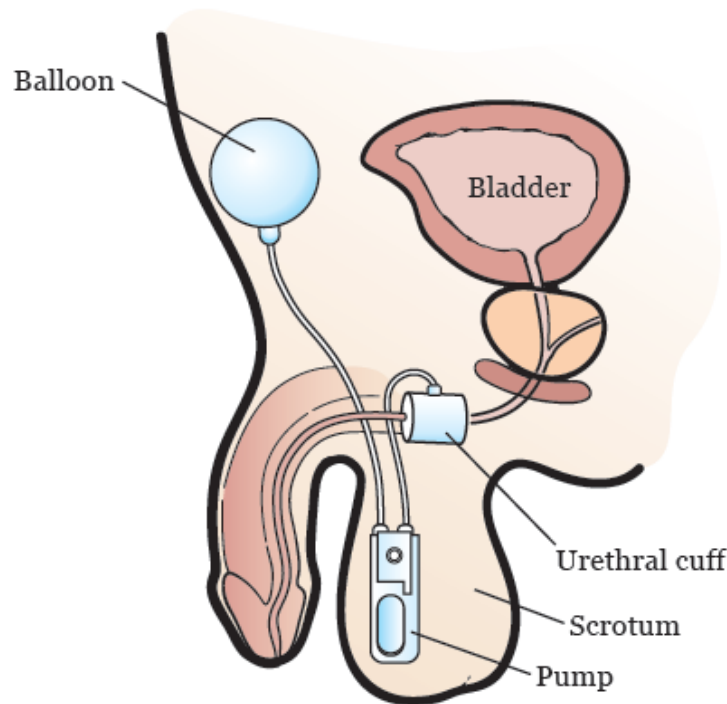


Figure 1: Schematic illustration of an artificial urinary sphincter (AUS) system, which consists of three main components: a urethral cuff placed around the urethra, a pressure-regulating balloon implanted in the abdomen, and a control pump located in the scrotum. [5]

1.3 Advantages of the New Mechanical Intraurethral Artificial Urinary Sphincter

To overcome the limitations of existing AUS devices, recent research has introduced a novel concept for a purely mechanical intraurethral artificial urinary sphincter. This design operates solely on mechanical and fluid dynamics principles, requiring no electronic components, thereby avoiding the need for external power supply, reducing the risk of failure, and minimizing maintenance requirements [7].

The core components include:

- Occlusion element (balloon or capsule): Positioned inside the urethra, normally filled with liquid to press against the inner wall of the outer cylinder and block urine flow
- Reservoir: Connected to the occlusion element via fine tubing, used to absorb and release fluid to regulate closing pressure
- Automatic pressure regulation: When bladder pressure rises briefly (e.g., due to coughing or sneezing), part of the fluid in the occlusion element temporarily flows into the reservoir to avoid excessive pressure and potential tissue damage. Once the pressure subsides, the fluid returns to the occlusion element to restore the seal

The overall structure of the system is illustrated in Figure 2. It demonstrates the internal connection between the occlusion element and the external reservoir, as well as the pressure regulation pathway.

The system offers the following advantages:

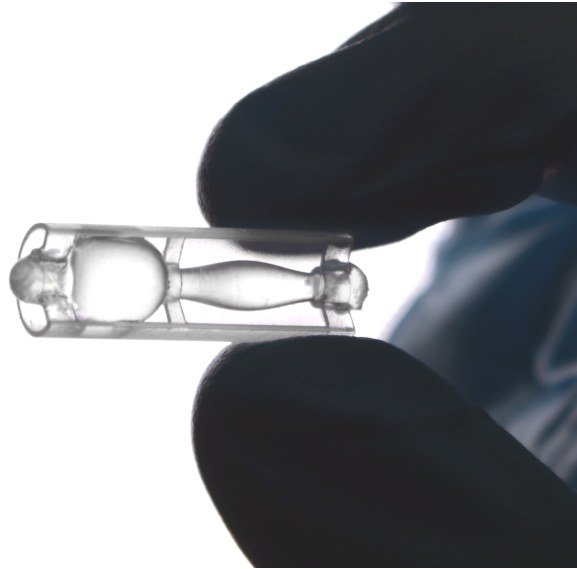


Figure 2: Cross-sectional view of the new mechanical intraurethral artificial urinary sphincter. The illustration shows the occlusion element positioned inside the urethra and connected via a fine tube to the external reservoir for automatic pressure regulation.

- No electronic control, avoiding risks of circuit failure.
- Simple structure, resulting in lower manufacturing and maintenance costs.
- Applicability to both male and female patients due to compact design.
- Minimally invasive implantation in an outpatient setting, avoiding complex multi-cavity surgery and enabling faster recovery.
- Intuitive operation: controlled naturally by abdominal pressure changes, without the need for manual activation.

The structural design of this system aims to enhance durability, improve ease of use, and reduce the need for long-term maintenance and additional surgical intervention.

1.4 Impact of Biofilm on Implants and Motivation for Test Platform Design

Implants that remain in continuous contact with urine, such as the new purely mechanical intraurethral AUS described earlier, face a significant risk of biofilm formation [8]. Biofilm, a layer of microorganisms embedded in a protective matrix, can impair device performance, shorten its lifespan, and increase the risk of clinical complications [8, 9, 10].

To address this challenge, this study focuses on developing an automated test platform to evaluate the device under realistic and long-term conditions. The platform is designed to:

- Simulate realistic micturition cycles and environmental conditions.
- Monitor the device's operational stability over extended periods.
- Observe and assess the effects of biofilm formation on the device's surface.
- Provide data for design improvement, durability enhancement, and pre-clinical validation.

1.5 Research Question and Objectives

This study aims to address the following core question:

Can an Arduino-based automated test bench, combined with a touchscreen user interface, be designed to simulate realistic micturition cycles, continuously monitor environmental temperature and UV LED status, and be used to analyze long-term biofilm formation on implants in a urine environment?

To answer this question, the following specific objectives are defined:

1. **User Interface (UI) Development:** Develop a touchscreen UI using the Arduino Giga Display Shield and the LVGL graphics library, allowing the user to set the number of daily micturition cycles, cycle duration, and start the system, while displaying environmental temperature, selected parameters, elapsed time, cycle count, and the status of the pump and UV LED.
2. **Control Software Development:** Implement control software using the Arduino Giga board and Motor Shield to operate the UV LED sterilization module and the peristaltic pump, and integrate a BMP280 sensor to measure environmental temperature.
3. **Automated Simulation:** Configure the system to automatically perform multiple micturition cycles per day, keeping the UV LED on during operation to simulate the sterile environment of the bladder prior to urination.
4. **Usability Testing:** Evaluate the UI using Jakob Nielsen's Usability Heuristics [11] to ensure ease of learning, operational efficiency, and error prevention, and verify core functionality against the specified use cases.

Scope. This thesis focuses on the design and verification of the automated test platform and its UI/control software. Clinical performance studies of the implant are out of scope and are considered only through bench-testing proxies.

Structure of the thesis. The remainder of this thesis is organized as follows: Chapter 2 reviews related work on biofilm formation, test benches, and user interfaces for embedded systems. Chapter 3 presents the system requirements and design specifications. Chapter 4 describes the implementation of the UI and control software. Chapter 5 reports the evaluation results, including usability and functionality testing. Chapter 6 critically analyze what the prototype achieves and what it can't yet do. Finally, Chapter 7 concludes the thesis and outlines potential directions for future work.

2 Foundations and state of the art

This chapter provides the scientific and technical foundation necessary to understand the development of the proposed automated test platform. It reviews the current knowledge of SUI, including its physiological mechanisms and epidemiology, and analyzes the limitations of existing AUS as well as the advantages of new mechanical sphincter designs. The mechanisms and risks of biofilm formation are discussed, with particular emphasis on the challenges of evaluating implants under realistic conditions. Furthermore, the chapter addresses the differences between static and dynamic biofilm testing environments, the role of embedded systems in biomedical device development, and principles of user interface and user experience (UI/UX) design in medical devices, including the use of the LVGL graphics library for embedded displays. This background will serve as the basis for defining the system requirements and design specifications in Chapter 3.

2.1 Medical background of SUI and its physiological mechanisms

SUI is defined as the involuntary leakage of urine during activities that increase intra-abdominal pressure, such as coughing, sneezing, laughing, or lifting heavy objects [3]. This condition occurs when the pressure inside the bladder exceeds the closing pressure of the urethra, resulting in an involuntary release of urine [1]. The underlying mechanism is often related to urethral sphincter insufficiency, pelvic floor muscle weakness, or impaired neural control of the urinary continence mechanism [3]. From a physiological perspective, urinary continence depends on the coordinated function of the bladder, urethra, pelvic floor muscles, and the central and peripheral nervous systems. The urethral sphincter and its surrounding supportive structures maintain closure pressure to prevent urine leakage during increases in intra-abdominal pressure. Damage or weakening of these structures, often due to childbirth, pelvic surgery, aging, or neurological disorders, can compromise this mechanism and lead to SUI [12].

SUI is estimated that over 420 million adults worldwide are affected by urinary incontinence, with SUI being the most common subtype among women [13]. In Germany, approximately 8–10% of the adult population is affected by urinary incontinence, with stress incontinence accounting for the majority of cases. Women are more than three times as likely to be affected compared to men, primarily due to anatomical differences and the impact of pregnancy and childbirth on pelvic floor integrity [1].

Clinically, SUI significantly affects quality of life, leading to social embarrassment, reduced physical activity, and psychological distress. In addition, it poses an economic burden on healthcare systems due to the costs of diagnosis, management, and potential surgical intervention. First-line treatments often include pelvic floor muscle training and lifestyle modifications, while more severe or refractory cases may require surgical options such as sling procedures or AUS implantation [14]. These interventions aim to restore adequate urethral closure pressure and improve continence, though each has its own limitations and risks.

2.2 Limitations and failure modes of existing AUS devices

AUS have been the gold standard for treating severe SUI for several decades. The conventional AUS consists of three main components: a cuff that encircles the urethra, a pressure-regulating balloon, and a control pump. The patient operates the device manually by squeezing the pump to temporarily deflate the cuff, allowing urination, after which the cuff automatically refills to restore continence.[5, 6]

Clinical studies have documented multiple limitations of existing AUS designs. High revision rates remain a major concern, with long-term follow-up showing that a substantial proportion of devices require replacement within a decade, primarily due to mechanical malfunction, infection, or tissue erosion [6]. Implantation is inherently invasive, requiring general or spinal anesthesia and postoperative hospitalization, which increases procedural risk in elderly or frail patients. Furthermore, the manual pump operation can be challenging for users with reduced hand strength or dexterity, and the majority of designs are optimized for male anatomy, limiting their applicability in female patients. These factors collectively underscore the need for more durable, less invasive, and user-friendly alternatives.

In addition to these limitations, device failure modes have been extensively reported. Mechanical malfunctions may involve leakage from the cuff or tubing, malfunction of the pressure-regulating balloon, or wear and breakage of pump components. Biological complications include device infection, urethral erosion, or urethral atrophy, which may require surgical removal of the device. Table 1 summarizes the most frequently reported complications and failure modes of AUS devices.

Table 1: Common failure modes and complications of AUS from [2], Sections 2

Category	Complication / Failure mode
Perioperative complications	Hematoma
	Pump migration
	Fluid loss
	Urinary tract infection
	Urinary retention
Late Post-Operative Complications	Device infection
	Urethral erosion
	Urethral atrophy
	Device Malfunction (PRB failure, cuff leakage, pump/tubing wear)
	Device explantation

2.3 Advantages of new mechanical Intraurethral AUS

To overcome the limitations associated with traditional AUS, such as high revision rates, surgical complexity, limited applicability across sexes, and usability challenges [2, 6], recent research has explored the development of a purely mechanical intraurethral closure system. Preis et al. (2022) [7] propose a novel AUS concept that operates entirely without electronic components or external power sources, relying solely on mechanical and fluid dynamic principles for regulation.

As shown in Figure 3, this innovative system centers around a pressure-sensitive balloon enclosed in a cylindrical tube that is implanted into the urethra. The balloon is connected to a throttle and a compensating reservoir, forming a self-regulating pressure mechanism. When the bladder pressure rises during normal filling or sudden abdominal pressure (e.g., from coughing), the system remains sealed. Only during sustained voluntary abdominal pressure—such as intentional micturition—does the system allow fluid to flow from the balloon into the reservoir, reducing its volume and opening the urethral passage.

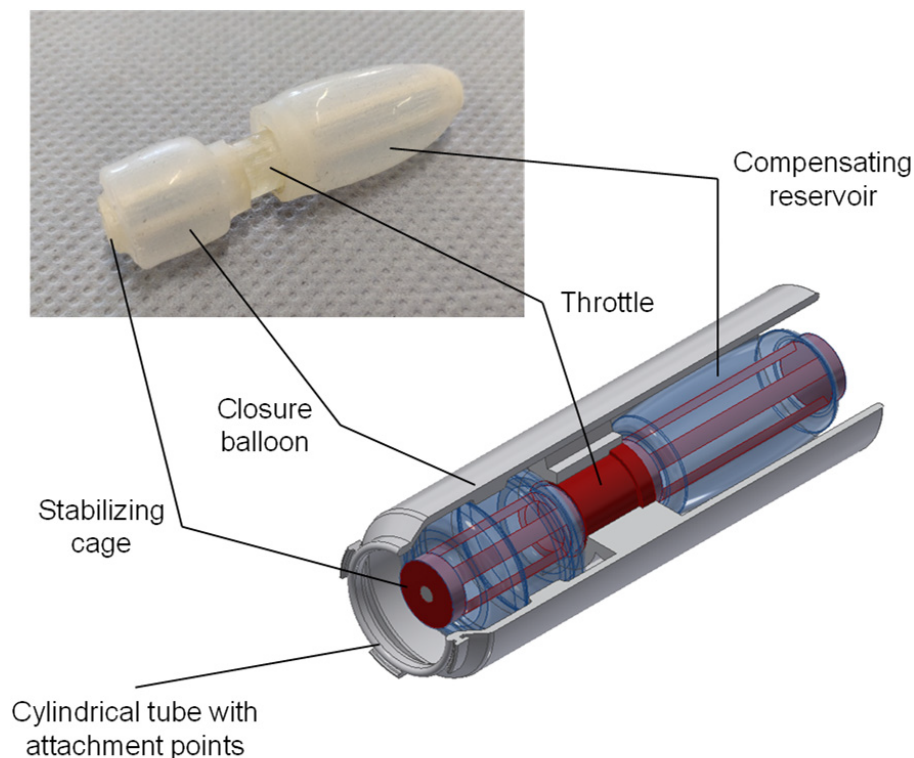


Figure 3: Prototype of the mechanical intraurethral artificial urinary sphincter [7]

The main advantages of this design are:

- No electronic components: This significantly reduces the risk of electronic failure, eliminates the need for power supply or recharging, and increases system reliability.
- Minimal implantation effort: The intraurethral design avoids invasive, multi-cavity surgery and enables simpler, outpatient-compatible procedures.
- Compact size and gender-neutral use: The smaller dimensions of the device allow for sex-independent application, addressing a key limitation in many existing AUS systems.

- Intuitive control mechanism: The system leverages natural bladder and abdominal pressure dynamics, requiring no manual activation and making it more user-friendly, especially for elderly patients or those with limited dexterity.
- Cost-effectiveness: A simplified mechanical structure is expected to lower both manufacturing and long-term maintenance costs.
- Higher expected durability: Fewer components and interfaces translate to improved mechanical robustness and a potentially lower revision rate.

Experimental data presented by Preis et al. demonstrated the system's capability to remain sealed during typical bladder filling and pressure spikes, while reliably opening under controlled conditions (i.e., sustained increased bladder pressure during voiding). In particular, a bladder pressure exceeding 100 cmH₂O sustained for more than 5 seconds was sufficient to initiate controlled opening, fulfilling the core functional requirement of continence support [7].

In summary, this new design addresses multiple drawbacks of conventional AUS systems and represents a promising direction for future patient-friendly and minimally invasive treatment of stress urinary incontinence.

2.4 Biofilm formation: mechanisms, risks, and clinical implications

Biofilms are complex communities of microorganisms embedded in a self-produced extracellular polymeric substance (EPS) matrix that adhere to both biotic and abiotic surfaces [9]. These structures pose a significant threat to the performance and longevity of implantable medical devices, particularly those continuously exposed to bodily fluids such as urine [15, 16]. As illustrated in Figure 4, the biofilm development process comprises five main stages: (1) reversible attachment, (2) microcolony formation, (3) maturation I, (4) maturation II, and (5) dispersion [17]. This model refines earlier simplified three-step concepts by distinguishing specific phases of growth and structural reorganization. Each stage involves unique biological mechanisms and cellular behaviors that enhance the biofilm's complexity, stability, and resistance to treatment.

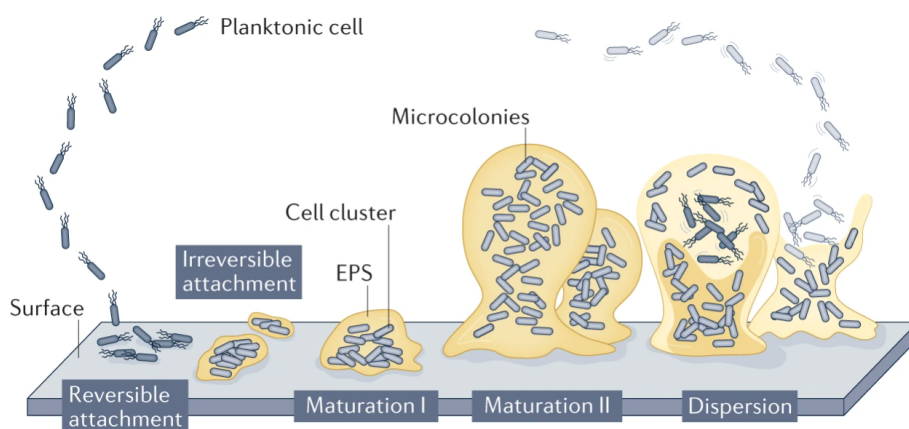


Figure 4: The five-step model of biofilm development [17]

- Attachment: Initially, planktonic (free-floating) bacteria adhere to a surface through weak interactions. This step is reversible and influenced by surface properties such as hydrophobicity and charge.
- Attached cells begin to divide, forming small clusters. During this stage, EPS production begins, and intercellular communication through quorum sensing increases.
- The biofilm structure becomes more complex. Layers of cells and EPS form three-dimensional architectures with channels for nutrient flow. Cells exhibit increased resistance to antibiotics and immune system attack.
- Some bacteria revert to a planktonic state and detach from the biofilm. This enables colonization of new surfaces and the formation of new biofilms.

In the context of urinary implants, this biofilm lifecycle is particularly problematic. Donlan (2002) [9] reported that biofilms can form within hours on polymer surfaces exposed to fluid environments, with full maturation occurring within several days. The urine-filled bladder is an ideal environment for biofilm development due to warmth, consistent moisture, and nutrient presence. Flores-Mireles et al. (2015) [18] further highlighted that uropathogens, such as *Escherichia coli* and *Proteus mirabilis*, are capable of forming persistent biofilms on devices like catheters and artificial sphincters.

Biofilm formation can lead to several device-related complications:

- Mechanical obstruction due to bacterial mass and EPS accumulation.
- Infection risk, including catheter-associated urinary tract infections (CAUTIs) and chronic inflammation.
- Increased revision surgery, as mature biofilms are difficult to eradicate without device removal.

Furthermore, traditional in vitro models used for biofilm testing, such as static culture plates, fail to replicate the dynamic physiological conditions found in the urinary tract. These systems often lack flow, temperature control, or cyclic pressure variations, which are essential to simulate realistic use scenarios. Xiong et al. (2021) [19] reviewed dynamic bladder models and emphasized that dynamic testing is more representative of the true biofilm formation process on urinary devices.

2.5 Comparison of Static and Dynamic Biofilm Testing Environments

Understanding the limitations of different biofilm testing environments is crucial when designing an appropriate in vitro platform for implant evaluation. The development and validation of anti-biofilm medical devices rely heavily on laboratory testing environments that can simulate real-life physiological conditions. In the field of biofilm research, both static and dynamic models have been widely used, each with distinct advantages and constraints depending on the research objective.

2.5.1 Static Biofilm Models

Static models are typically characterized by a fixed volume of nutrient medium without continuous flow. Examples include microtiter plate assays, colony biofilm models, and tube reactors. These models are simple, low-cost, and easy to handle, making them suitable for high-throughput screening or preliminary biofilm formation studies [19]. However, static conditions inherently lack the hydrodynamic forces, nutrient gradients, and shear stress found in physiological environments. As a result, biofilms grown under static conditions tend to exhibit less structural complexity and may not accurately replicate the behavior of clinical or device-associated biofilms.

2.5.2 Dynamic Biofilm Models

In contrast, dynamic testing environments are designed to simulate physiological processes more accurately. Dynamic models are particularly important for studying medical devices that remain in contact with bodily fluids, such as urinary catheters or intraurethral implants. These systems incorporate fluid circulation, programmable pressure modulation, and temperature control to replicate *in vivo* conditions. According to Xiong et al. (2021), dynamic bladder models are better suited for studying catheter-associated urinary tract infections (CAUTIs) and biofilm development under flow conditions. Dynamic models can simulate:

- Urine flow rate and direction
- Periodic urination cycles
- Shear stress acting on the device surface
- Nutrient replenishment and waste removal

Moreover, studies have shown that bacterial adhesion, biofilm structure, and resistance to antimicrobials differ significantly between static and dynamic models [20]. For instance, dynamic flow can influence biofilm thickness, architecture, and detachment behavior, which are critical factors for evaluating the long-term reliability of urinary implants.

The absence of standard dynamic models for testing intraurethral sphincter devices represents a key gap in current research infrastructure. Existing systems are often designed for catheters or general-purpose surface testing, and do not accommodate device-specific factors such as internal occlusion mechanisms or abdominal pressure simulations. A prominent example is the modified Robbins device (MRD), which exposes multiple surface coupons to controlled flow and has been widely used to evaluate biofilm formation on different materials and to test antibiotic lock therapies under dynamic conditions [21]. This limitation further emphasizes the need for a custom-designed, automated test bench that can simulate urination cycles, environmental temperature, and UV exposure while providing continuous monitoring.

The proposed test platform in this thesis aims to fill this gap by integrating programmable cycle control, peristaltic pumping, environmental sensors, and a touchscreen interface. This will enable repeatable, long-term biofilm testing in conditions that closely approximate the human urinary environment, improving the translational relevance of preclinical evaluations.

2.6 Embedded systems in biomedical device development

Embedded systems play a critical role in the development of modern biomedical devices due to their compact size, modularity, and real-time responsiveness. These systems are designed to meet demanding clinical requirements such as continuous monitoring, automated control, and accurate data collection. With the advancement of microcontroller (MCU) technologies and low-power components, it has become feasible to integrate sensing, computation, and user interaction into a single, intelligent medical subsystem [22].

As described by Arandia et al. (2022) [22], embedded systems in healthcare typically exhibit the following key characteristics:

- **Modularity:** Allowing the system to be customized or extended for various medical scenarios by integrating sensing, control, communication, and display modules.
- **Real-time responsiveness:** Ensuring immediate reaction to physiological changes—for example, regulating fluid flow, triggering alarms, or adjusting device parameters.
- **Data handling:** Capable of storing and pre-processing sensor data for trend analysis, device tracking, or assisting clinical decisions.
- **Low power consumption and long-term reliability:** Especially important for devices that must operate continuously in body-contact environments, such as implants or catheter systems.

In this study, the embedded system serves as the core architecture of the automated test bench for evaluating a novel intraurethral bladder assist device. The system is based on the Arduino platform and consists of the following modules:

- **Sensor unit:** A BMP280 sensor continuously monitors ambient temperature to simulate physiological test conditions.
- **Actuation unit:** A peristaltic pump simulates urination cycles, while a UV LED module provides sterilization throughout the cycle.
- **User interface unit:** The Arduino Giga Display Shield, combined with the LVGL graphics library, provides a touchscreen interface for configuring test parameters and displaying real-time system status.
- **Control logic unit:** A microcontroller (Arduino) executes automated timing and operational logic to perform multiple urination cycles per day without manual intervention.

This architecture reflects the modular structure suggested by Arandia et al., enabling flexible development and verification at the component level. Moreover, by using off-the-shelf hardware and open-source software tools, this setup supports rapid prototyping while maintaining a clear separation between hardware abstraction and medical function logic.

Although this prototype is not intended for clinical deployment, aligning with the design principles discussed by Arandia et al. [22] ensures that future iterations can be developed toward regulatory compliance.

2.7 Principles of UI/UX in medical devices and use of LVGL in embedded displays

User interface (UI) and user experience (UX) design are crucial in the development of medical devices, where intuitive operation, error prevention, and reliability directly impact user safety and device effectiveness. Unlike consumer products, medical device interfaces must be designed with careful consideration of clinical context, user roles (e.g., patients, nurses, technicians), and stress levels during operation. Poor UI design may result in operational errors, reduced usability, or even adverse clinical outcomes [23].

2.7.1 Jakob Nielsen's 10 Usability Heuristics for User Interface Design

To address these challenges, this project adopts Jakob Nielsen's 10 Usability Heuristics for User Interface Design, which provide a structured foundation for evaluating and designing effective interfaces [11]. The following key principles were applied during the development of the embedded UI for the bladder assist device test platform:

1. **Visibility of system status:** A good system should always keep the user informed about what is happening. This can be done by showing updates, progress bars, or system status so that users know if the system is working or waiting for input.
2. **Match between system and the real world:** The interface should use simple and familiar language. Icons, terms, and functions should reflect real-life meanings, so users can easily understand what each function does without needing technical knowledge.
3. **User control and freedom:** Users should be able to easily undo or stop an action if something goes wrong. Functions like "cancel," "back," or "pause" are important to help users feel safe when using the system.
4. **Consistency and standards:** The same words, colors, icons, and layout styles should be used across the whole system. If something means "start" on one page, it should mean the same on all others. This helps users learn the system faster and avoid mistakes.
5. **Error prevention:** Instead of showing error messages after something goes wrong, the system should try to prevent errors before they happen. For example, by limiting input ranges, using drop-down menus, or reminding users to complete all required settings.
6. **Recognition rather than recall:** Users should not be forced to remember commands or values. All the important options, values, and actions should be visible on the screen. This reduces cognitive load and makes the system easier to use.
7. **Flexibility and efficiency of use:** The system should support both new users and experienced users. Beginners may need simple buttons, while experts may prefer shortcuts or advanced settings to finish tasks faster.
8. **Aesthetic and minimalist design:** The design should be clean and not include too much information or decoration. Too many elements can distract users and make the interface harder to use. Every detail on the screen should have a clear purpose.

9. Help users recognize, diagnose, and recover from errors: When an error happens, the system should show a helpful message to explain what went wrong and how to fix it. The message should be simple and guide users step-by-step.
10. Help and documentation: Even though the system should be easy to use without instructions, some users may still need help. The system should offer clear and accessible help, like tooltips, short instructions, or basic user guides.

2.7.2 Benefits of LVGL in Embedded Systems

To implement these UI principles in the embedded environment, the LVGL (Light and Versatile Graphics Library, version 9.2) [24] was used in combination with the Arduino Giga Display Shield. LVGL provides a lightweight yet powerful framework for building modern graphical interfaces on resource-constrained hardware.

Key advantages of using LVGL in this project:

- Cross-platform support: Compatible with a wide range of MCUs and displays, making it ideal for Arduino-based systems.
- Touch-friendly widgets: Includes ready-to-use components such as sliders, buttons, charts, and labels, optimized for touchscreens.
- Custom styling: Allows customization of colors, fonts, and layouts to align with medical UI aesthetics.
- Low memory footprint: Efficient rendering engine designed for embedded devices with limited RAM and flash memory.
- Real-time performance: Supports smooth animations, status updates, and user interactions without noticeable delay.

In this project, LVGL was used to implement a modular UI structure with separate tabs for configuration, real-time monitoring, and system status. This design ensures clarity and supports long-term usability evaluation.

Overall, the integration of Nielsen's usability heuristics with LVGL enables a robust and user-centered interface for the test platform, supporting safe and efficient interaction even under repeated or long-term use in laboratory simulations.

3 Need for action

This chapter defines the scientific and engineering rationale for developing an automated test platform capable of evaluating intraurethral artificial urinary sphincters under physiologically relevant conditions. It identifies the methodological and functional limitations of current biofilm and mechanical testing approaches, emphasizing the absence of systems that can reproduce cyclic urination and continuous urine exposure. Building on the literature analysis presented in Chapter 2, the discussion highlights the need for dynamic, reproducible, and device-specific evaluation environments. The chapter further derives the core system requirements from literature, translating them into measurable design objectives. Finally, it outlines the conceptual framework of the proposed Arduino-based platform, describing its embedded control architecture, sensing and actuation modules, UV-LED suppress microbial growth concept, and touch-screen graphic user interface designed with the LVGL graphics library.

3.1 Current Limitations of Existing Test Systems

Research on urinary implants and AUS increasingly recognizes the necessity of pre-clinical environments that can mimic physiological conditions of the lower urinary tract. However, most current testing platforms remain limited to simplified mechanical or microbiological assessments that fail to capture the dynamic interaction between pressure cycles, fluid exposure, and surface biofilm development [8]. Existing mechanical rigs are typically designed to evaluate basic flow resistance or component wear, while microbiological assays focus on early bacterial adhesion on static substrates [9, 25]. As a result, neither approach adequately represents the combined physical and biological stresses acting on intraurethral implants during prolonged use.

3.1.1 Static test models

Most in-vitro biofilm evaluations are performed under static conditions, typically using microtiter plates, colony biofilms on agar, or glass-slide system [8]. These techniques provide reproducible conditions for short-term bacterial attachment but lack physiological relevance. Without continuous medium renewal or cyclic flow, the in-vitro environment remains compositionally uniform, so nutrients and oxygen do not redistribute realistically and biofilm stratification and maturation are hindered [8, 26]. In addition, static setups cannot reproduce the intermittent flow and pressure fluctuations that accompany bladder filling and voiding [19]. Consequently, static biofilms often exhibit thinner matrices, reduced extracellular polymeric substance (EPS) production, and lower resistance to detachment compared to those found on explanted devices [26]. Although such models are valuable for preliminary screening, they provide limited insight into long-term implant–biofilm interactions.

3.1.2 Dynamic flow and catheter models

To address these shortcomings, several dynamic bladder and catheter systems have been developed [19]. In addition, general-purpose flow-cell systems such as the MRD, introduced

in Section 2.5.2, enable reproducible biofilm growth on multiple coupons under continuous flow and have been widely used to evaluate surface-modified materials and disinfection or lock strategies [21]. Dynamic bladder and catheter models typically circulate artificial urine or human urine and allow repeated assessment of microbial growth under flow [19]. While these systems represent an important step toward physiological simulation, most do not implement a fully closed, pressure-regulated configuration. As a result, flow patterns are often unidirectional and approximately constant, whereas the human urinary tract exhibits cyclic pressure changes and periods of stasis [19, 27]. Moreover, existing dynamic platforms generally focus on microbiological or material endpoints and only rarely include integrated sensing for device-status feedback, which makes it difficult to correlate mechanical parameters with biological outcomes in real time [19]. Studies have emphasized that variations in shear exposure, retention time, and pressure oscillation critically influence microbial adhesion and detachment kinetics [8, 25]. In the absence of sufficiently resolved environmental monitoring, these effects remain poorly quantified.

3.1.3 Mechanical and functional evaluation

Existing mechanical evaluation systems for AUS do not account for biochemical processes such as biofilm accumulation or material degradation over time. For example, clinical evidence shows that even well-functioning AUS devices can develop microbial layers that alter local pressure transfer and valve behavior, contributing to mechanical malfunction or urethral erosion [2, 10]. Therefore, assessing long-term reliability requires test benches that integrate both mechanical cycling and biological exposure. The absence of such combined evaluation platforms remains a major bottleneck in translating novel AUS concepts into clinical practice.

In summary, current test systems suffer from three main limitations:

1. Static assays that lack physiological dynamics
2. Dynamic flow models that do not reproduce cyclic pressure regulation
3. Mechanical setups that omit biological interaction and long-term degradation factors

These deficiencies underline the need for an integrated, automated test platform capable of reproducing realistic urination cycles, maintaining environmental control, and enabling continuous observation of device performance under biofilm-forming conditions. Such a system would bridge the methodological divide between mechanical endurance testing and biological evaluation, creating a unified framework for long-term, physiologically relevant device assessment.

3.2 Technical and Clinical Motivation

The motivation for developing an improved pre-clinical evaluation platform for AUS arises from the persistent clinical complications and design challenges reported over decades of use. Clinically, revision and explantation rates remain significant, often due to infection, erosion, or urethral atrophy caused by long-term interaction between the implant and the surrounding tissue [2, 4]. These outcomes reflect not only surgical and anatomical factors but also the lack of pre-clinical methods that can predict device behavior under realistic urinary conditions. From

an engineering perspective, the main challenge lies in controlling fluid dynamics, implementing cyclic actuation, and characterizing the surface changes induced by microbial activity, all of which collectively affect device performance over time. Bridging this knowledge gap requires a system that enables controlled, repeatable simulation of these interactions in a measurable way.

3.2.1 Clinical perspective

AUS has been considered the gold-standard surgical treatment for severe or refractory cases for more than four decades [2]. Despite its therapeutic value, modern studies continue to report substantial long-term revision rates, ranging between 20% and 30% within ten years of implantation, largely due to mechanical failure, infection, or tissue erosion [2, 4]. Clinical analyses of explanted devices have further demonstrated that biofilm formation has been frequently observed among indwelling AUS, even in asymptomatic patients [10]. Such biofilms may alter local pressure transmission, impede cuff deflation, and initiate chronic inflammation, ultimately reducing the device's lifespan [2]. These outcomes highlight that current pre-clinical testing fails to predict many failure mechanisms encountered in vivo. To improve clinical reliability, bench-testing must therefore replicate the cyclic pressure exposure and biological interactions that occur inside the urinary tract.

3.2.2 Technical perspective

From a technical standpoint, the challenge lies in bridging mechanical precision with biological realism. In clinical reality, mechanical cycling and microbial activity coexist, and their combined effects are not captured by most in-vitro models that rarely incorporate simultaneous mechanical or physiological stresses [8, 25].

Recent progress in embedded systems and open-source microcontroller technology now provides an opportunity to overcome these limitations. Platforms such as the Arduino Giga combine multiple input/output channels, high-precision timing, and compatibility with advanced graphical libraries like LVGL, enabling the integration of sensing, control, and user interaction within a compact framework [22, 24]. When applied to biomedical testing, such systems allow real-time environmental monitoring, automated actuation, and long-term data logging—capabilities essential for reproducing physiological conditions in laboratory settings.

3.2.3 Bridging both domains

The convergence of clinical and technical demands forms the central motivation for the present work. Clinicians require a better understanding of how device design interacts with biological processes to inform preventive maintenance and reduce revision rates, while engineers need reproducible, data-driven tools to validate new prototypes before animal or human studies. Developing an automated, sensor-integrated test bench represents a practical and scalable approach to meeting both objectives. It enables continuous operation under physiologically relevant conditions, quantitative tracking of environmental variables, and structured evaluation of long-term device behavior. By linking engineering validation directly to clinical relevance, such a system lays the foundation for more reliable AUS designs and, ultimately, improved patient outcomes.

3.3 Proposed Concept and Objectives

The insights gained from previous sections underline the necessity of an experimental platform that can integrate mechanical simulation, environmental monitoring, and biological exposure within a single automated framework. The proposed concept builds upon the principles of reproducibility, modularity, and physiological relevance, using an embedded control system to emulate the functional environment of an intraurethral AUS.

3.3.1 System concept and Design objectives

The test bench is designed around a microcontroller-based architecture that coordinates sensing, actuation, and data logging in real time. The Arduino Giga board, combined with the Arduino Motor Shield Rev3 and Arduino Giga Display Shield, serves as the central control unit, providing multiple I/O interfaces, high-precision timing, and full compatibility with advanced display and control libraries [28, 29, 30]. A peristaltic pump, operated through the Motor Shield, produces adjustable flow cycles to simulate the temporal characteristics of bladder filling and urination cycles under controlled laboratory conditions. Environmental parameters, including temperature, are continuously monitored by a BMP280 sensor [31], ensuring stable and reproducible test conditions throughout operation.

The control software is developed in Arduino language based on C/C++ within the Arduino IDE, implementing a structured loop that performs periodic cycle execution, sensor acquisition, and data transmission [32]. A touchscreen graphical user interface, developed with the LVGL graphics library [24], allows users to configure experimental parameters (e.g., cycle duration, number of repetitions, and flow rate), visualize real-time system status, and export logged data. The interface follows Jakob Nielsen's usability heuristics, ensuring that the system remains intuitive, transparent, and resistant to operator error [11, 24].

The conceptual design of the platform aims to achieve the following objectives:

- Physiological realism: Reproduce urination and filling cycles under controlled laboratory conditions to mimic realistic intraurethral operation.
- Automation and reproducibility: Enable long-term, unattended operation with repeatable cycle parameters and automated environmental recording.
- Integrated sensing: Continuously track temperature and time, linking environmental fluctuations to observed mechanical or biological outcomes.
- User-centered control: Provide a touchscreen interface for direct parameter manipulation, visual feedback, and error prevention through interactive cues.
- Data traceability: Record environmental temperature over fixed time intervals for subsequent analysis.

3.3.2 Scientific relevance

By merging mechanical and biological evaluation within a single automated system, the conceptual design of this study directly addresses the methodological gap identified earlier. It provides a dynamic yet controllable environment for investigating how biofilm formation and environmental factors interact to affect device longevity. The Arduino-based implementation ensures affordability, flexibility, and accessibility for both academic and clinical laboratories.

Furthermore, the integration of usability engineering principles introduces a human-factor dimension in experimental instrumentation, aligning laboratory design with clinical-grade operability [11].

3.4 Implementation Strategy and Core Functional Modules

The automated test platform is structured around four core functional units, each responsible for a specific operational task, collectively enabling dynamic simulation, environmental monitoring, and intuitive user control. These units cover functions ranging from fluid circulation and environmental data acquisition to interface usability and operational safety, allowing the system to perform long-term testing under stable conditions while ensuring reproducible results.

3.4.1 Graphical User Interface (GUI) Control

The graphical interface constitutes the operational core of human–machine interaction. Built with the LVGL graphics library (version 9.2) on the Arduino Giga Display Shield, it provides a structured environment for parameter input, and real-time visualization. All experimental variables—such as cycle duration, flow rate, temperature, and repetition count—can be configured directly on screen through slider and selector widgets. The event-driven rendering system of LVGL enables smooth transitions, efficient refresh cycles, and low memory consumption, making it well suited for embedded biomedical applications [24].

3.4.2 Simulation Cycle Module

To replicate the functional dynamics of the urinary tract, the simulation module employs a peristaltic pump controlled by the Arduino Giga's PWM and timing functions via the Motor Shield Rev3. By adjusting cycle duration and resting intervals, the system emulates alternating bladder-filling and voiding phases. This cyclic pattern reproduces the human urination process and is crucial for evaluating the interaction between mechanical and biological factors under repeated operation [19, 8]. In addition, a UV-LED module, also driven through the Motor Shield Rev3, maintains a controlled sterility environment within the test vessel.

Continuous cyclic operation over several days to months enables investigation of biofilm attachment and detachment behavior within the urinary tract. Compared with static or unidirectional flow systems, this cyclic configuration produces more physiologically relevant data, providing a reliable foundation for the long-term evaluation of AUS devices.

3.4.3 Device Status Monitoring

The platform integrates a BMP280 sensor to continuously measure ambient temperature and record the last 20 data points relative to the current time. These values are visualized in real time through a temperature trend graph, allowing users to easily monitor environmental stability and temporal variations. In addition to environmental monitoring, the system automatically logs operation time and cycle count, ensuring that each testing phase includes a precise and traceable timestamp. On the display interface, the real-time operational status of all major

components, including the pump and UV LED module, is represented through graphical icons, enabling users to instantly verify device activity and system response. Furthermore, the data structure includes a reserved humidity field, allowing seamless future integration of additional environmental sensors (e.g., DHT22 or BME280) to further enhance environmental control and monitoring capabilities.

3.4.4 Usability Testing Based on Nielsen's Heuristics

Beyond technical validation, the platform also includes a dedicated usability evaluation to assess operational efficiency and human-factor performance. The evaluation is conducted based on Nielsen's 10 usability heuristics [11], covering aspects such as system consistency, visibility of feedback, user control, and error-recovery capability. Each of the ten principles is systematically reviewed to ensure compliance. The evaluation results will help enhance the efficiency and reliability of human-machine interaction and provide a foundation for interface refinement and operational optimization.

Integrating usability evaluation into the development process follows the same principles adopted in medical device certification, ensuring that the system performs not only with mechanical stability but also with operational safety and intuitive usability [33]. Through this process, the platform evolves from a purely experimental apparatus into a user-centered research tool adaptable to both engineering and biomedical research environments.

Together, these functional modules establish an integrated framework that unites mechanical simulation, environmental control, and human-machine interaction. The system's modular structure allows continuous operation under reproducible conditions while remaining flexible for future expansion. By embedding usability, monitoring, and control within a single architecture, the implementation strategy directly addresses the methodological limitations discussed earlier and provides a foundation for future work presented in Chapter 4.

4 Methods

This chapter describes the technical implementation of the automated test platform developed to simulate urination cycles and analyze biofilm-related phenomena on implantable devices. It covers the hardware configuration, software architecture, and user-interface implementation, as well as the methods used for sensor feedback, cycle control, and usability testing. The study focuses on reproducibility and ensures clear and stable interaction between mechanical, electronic, and software components. The complete implementation is publicly available on GitHub [34].

4.1 System Architecture

The platform follows a modular embedded-system architecture combining Arduino Giga R1 WiFi (Figure 5), the Arduino Giga Display Shield (Figure 6), and the Arduino Motor Shield Rev3 (Figure 7). The Giga board provides a 32-bit microcontroller (STM32H747XI), and its abundant digital and analog I/O pins enable simultaneous connection of actuators and sensors without multiplexing. The Display Shield hosts a 4-inch capacitive touchscreen used for configuration and monitoring, while the Motor Shield serves as a dedicated motor-control interface, converting low-power control signals from the Giga board into stable, high-current outputs required to drive DC motors and pumps. It supports bidirectional speed and direction control through pulse-width modulation (PWM).

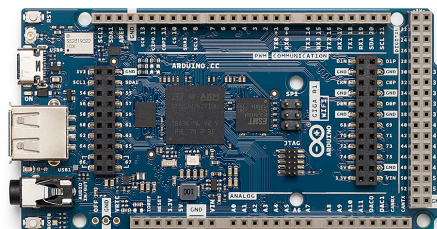


Figure 5: Arduino GIGA R1 WiFi [28]

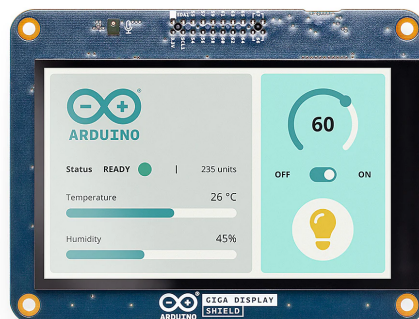


Figure 6: Arduino Giga Display Shield [29]

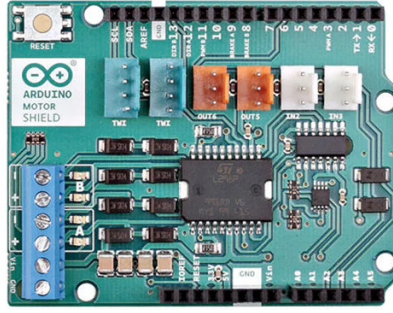
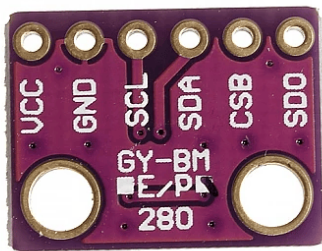


Figure 7: Arduino Motor Shield Rev3 [30]

The automated test bench incorporates three peripheral components, as illustrated in Figure 8:

1. BMP280 sensor: Installed inside the test chamber, which continuously records ambient temperature parameters (Figure 8 a))
2. Grothen G928 peristaltic pump: Responsible for generating controlled inflow and outflow during the simulated micturition cycles (Figure 8 b))
3. UV-LED Ring (custom-built): Mounted above the liquid container, which helps suppress microbial growth to extend the reusability of the test fluid (Figure 8 c))



a) BMP280 sensor [31]



b) Grothen G928 peristaltic pump [35]



c) Custom-built UV-LED ring

Figure 8: Peripheral components of the automated test platform: (a) BMP280 sensor, (b) Grothen G928 peristaltic pump, and (c) custom-built UV-LED ring

All sensor devices communicate via the I2C interface, while the actuation modules are controlled through PWM and digital outputs. The power supply is provided through the Arduino Giga Motor Shield's external AC adapter (input: AC 220 V, output: DC 12 V). The circuit distributes 9.4 V (78% PWM) to the peristaltic pump and 12 V to the UV-LED ring, respectively.

From a systems-integration perspective, the architecture implements a three-layer structure:

1. Hardware Layer: Includes pump, UV-LED, and sensors connected through dedicated motor-driver and I2C lines
2. Control Layer: The Arduino Giga R1 WiFi that executes cycle scheduling, actuator control, and sensor polling in real time

3. Interface Layer: The LVGL-based GUI running on the display shield, handling user input, visualization, and data logging

This architecture establishes a stable hierarchical relationship between functional allocation and information flow, making the control and feedback processes more predictable. Its overall coordination ensures measurement accuracy and operational consistency, providing a reliable foundation for subsequent functional validation and interface integration.

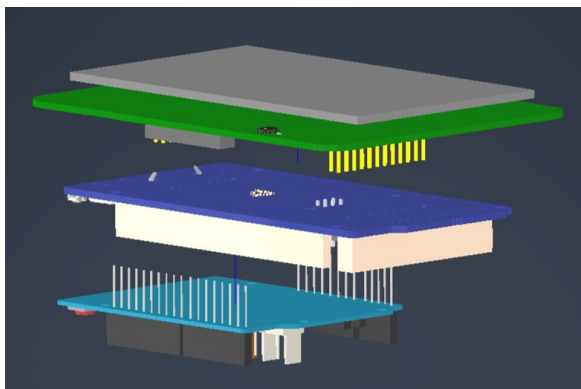
4.2 Component Setup and Hardware Integration

Each component described in Section 4.1 was physically integrated into a compact modular assembly. The Arduino Giga R1 WiFi is stack with the Display Shield and Motor Shield to reduce wiring complexity. The shields communicate directly via standard Arduino headers, while peripheral components such as the peristaltic pump, UV-LED module, and BMP280 sensor are connected through dedicated PWM and I2C lines.

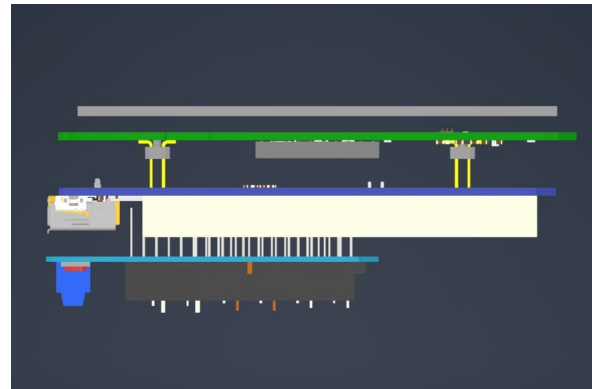
4.2.1 Control Core and Shield Stack

The entire system is centered around the Arduino Giga R1 WiFi board, which is equipped with an STM32H747XI dual-core processor (Cortex-M7 and Cortex-M4). This architecture allows clear task separation, the M7 core handles real-time control, sensor data acquisition, and the LVGL interface (including touch event processing). The M4 core manages display output and refreshing pre-rendered frames to the panel. This decoupling between “rendering and display transfer” and the time-critical actuation loops prevents GUI refresh operations from interfering with control timing, thereby ensuring stable long-term performance. [28]

The Display Shield and Motor Shield are connected via standard Arduino interfaces and directly stacked on the Giga main board, forming a compact vertical modular assembly. As shown in Figure 9, the stacking order from top to bottom is the Arduino Display Shield, the Giga R1 WiFi board, and the Motor Shield.



a) Exploded view



b) Side view

Figure 9: Exploded and side views of the integrated hardware stack showing the Arduino Giga R1 WiFi, Display Shield, and Motor Shield used in the test platform

4.2.2 Power Conversion and Distribution

Figure 10 illustrates the electrical architecture of the actuation subsystem, showing the power flow and wiring relationships among the AC adapter, Arduino Motor Shield Rev3, peristaltic pump, and UV-LED module.

The system receives electrical power from a dedicated AC adapter, which converts the 220 V AC mains into a stable 12 V DC supply. This 12 V input is directly connected to the Motor Shield's external power terminals, serving as the primary energy source for both the actuation components and the control electronics.

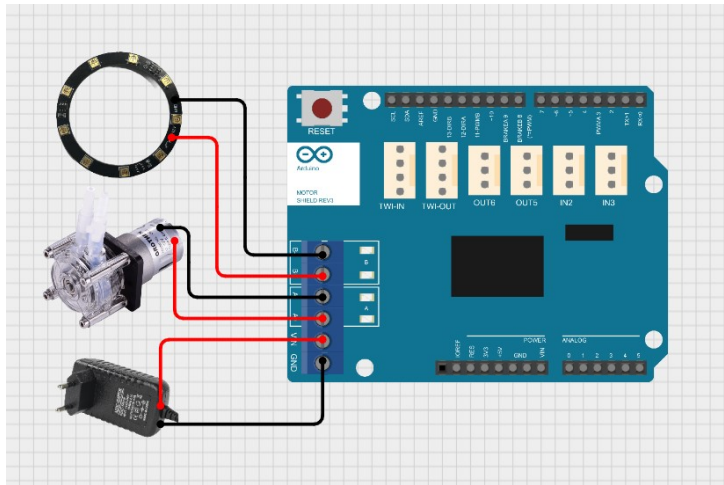


Figure 10: Electrical wiring of the actuation module, showing power flow from the 220 V AC adapter to the Motor Shield, pump, and UV-LED. The Motor Shield also supplies regulated 5 V power to the Arduino Giga R1 WiFi and Display Shield

The Motor Shield functions as a central power and control hub. It delivers regulated 9.4V and 12 V outputs to the peristaltic pump and UV-LED ring, while simultaneously supplying 5 V regulated power through the internal pin headers to the Arduino Giga R1 WiFi and Display Shield stacked above it. This configuration allows the entire system to be powered from a single AC adapter, simplifying wiring and ensuring synchronized startup behavior. This hierarchical design provides a unified and compact energy architecture, ensuring efficient and reliable performance of the automated test platform.

4.2.3 Actuation Module: Peristaltic Pump and UV-LED Control

The actuation module constitutes the core subsystem responsible for simulating fluid discharge and maintaining a clean experimental environment within the automated urinary test platform. It comprises a 12 V DC peristaltic pump and an LED ring module, both components are powered and controlled via the Arduino Motor Shield Rev3, while the Arduino Giga R1 WiFi board manages their activation logic through digital output pins. The firmware coordinates both devices according to user-defined parameters configured in the graphical user interface.

The peristaltic pump is connected to the Motor A terminals of the Motor Shield, whereas the LED ring is connected to Motor B (Figure 10). Each channel of the shield contains an H-bridge driver controlled by three digital pins — In1, In2, and ENA, where:

- In1 and In2: Determine the rotation direction of the motor (forward or reverse)
- ENA (Enable): Control the effective output voltage and motor speed through PWM

The PWM duty cycle defines the proportion of time the 12 V supply is applied to the motor per cycle. For example, in the command `analogWrite(motorENA, 200)`, the parameter value 200 (out of a maximum of 255) corresponds to roughly 78% duty cycle, resulting in an effective average output voltage of approximately 9.4 V. This allows fine-tuned speed control without requiring an external variable power supply.

These pins are initialized as outputs during system startup:

```

1  const int motorIn1 = 12;
2  const int motorIn2 = 9;
3  const int motorENA = 3;
4  const int ledIn1   = 13;
5  const int ledIn2   = 8;
6  const int ledENA   = 11;
7
8  pinMode(motorIn1, OUTPUT);
9  pinMode(motorIn2, OUTPUT);
10 pinMode(motorENA, OUTPUT);
11 pinMode(ledIn1,  OUTPUT);
12 pinMode(ledIn2,  OUTPUT);
13 pinMode(ledENA,  OUTPUT);

```

In this configuration, pins 12, 9, and 3 are assigned to control the peristaltic pump, whereas pins 13, 8, and 11 correspond to the UV-LED. This assignment ensures non-overlapping control channels for both devices and allows independent adjustment through software commands.

During each simulated micturition cycle, the Giga R1 controls the pump's direction and speed using these three pins. When In1 is set HIGH, In2 LOW, and `analogWrite(motorENA, 200)` is executed, the motor runs forward at about 78% of full power. After the programmed duration, the pump stops and remains inactive until the next micturition cycle begins:

```

1  void startMotor(bool forward = true) {
2      digitalWrite(motorIn1, forward ? HIGH : LOW);
3      digitalWrite(motorIn2, forward ? LOW : HIGH);
4      analogWrite(motorENA, 200);
5  }
6
7  void stopMotor() {
8      digitalWrite(motorIn1, LOW);
9      digitalWrite(motorIn2, LOW);
10     analogWrite(motorENA, 0);
11 }

```

The UV-LED ring uses a similar control structure but remains continuously active during testing with full power:

```

1  void startLED() {
2      digitalWrite(ledIn1, HIGH);
3      digitalWrite(ledIn2, LOW);
4      analogWrite(ledENA, 255);
5  }
6

```

```

7 void stopLED() {
8   digitalWrite(ledIn1, LOW);
9   digitalWrite(ledIn2, LOW);
10  analogWrite(ledENA, 0);
11 }

```

4.2.4 Environmental Monitoring (BMP280 Sensor)

Figure 11 illustrates the electrical configuration of the environmental monitoring subsystem, showing the interconnection between the Arduino Giga R1 WiFi, the Motor Shield, and the BMP280 sensor. The BMP280 is powered through the shared 3.3 V and GND rails provided by the Giga R1. The sensor communicates directly with the Giga R1 via the standard I2C bus (SDA = pin 20, SCL = pin 21), enabling continuous measurement of ambient temperature and pressure throughout automated testing cycles.

In the actual circuit, when the Giga R1 board is stacked onto the Motor Shield, both boards share the same 3.3 V and GND power rails through their standard Arduino header pins. As a result, the BMP280's VCC and GND lines are connected through the Motor Shield's pin terminals, while the SDA and SCL lines remain wired to the corresponding I2C pins on the Giga R1.

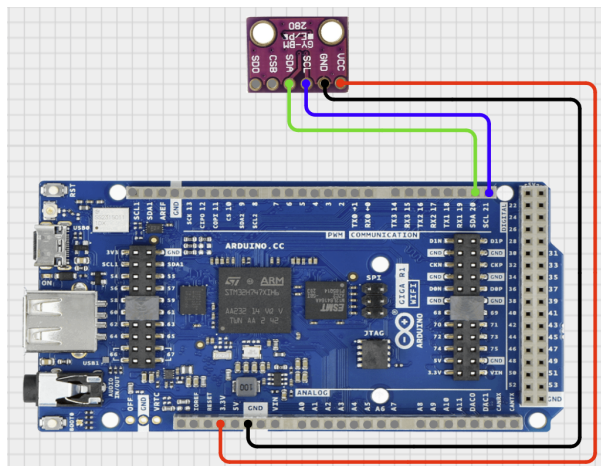


Figure 11: Schematic diagram of the BMP280 sensor connection to the Arduino Giga R1, showing wiring colors: VCC (red), GND (black), SCL (blue), and SDA (green)

The initialization procedure for the BMP280 sensor is shown below:

```

1 Adafruit_BMP280 bmp; // Create BMP280 sensor instance
2
3 void setup() {
4   Wire.begin(); // Initialize I2C communication (SDA=20, SCL=21)
5   if (!bmp.begin(0x76)) { // Check device at I2C address 0x76
6     Serial.println("BMP280_not_found");
7     bmp_error = true;
8   } else {
9     bmp_error = false;
10  }
11 }

```

At startup, `Wire.begin()` initializes the default I2C communication channel on pins 20 and 21, and the firmware performs a handshake at address 0x76 to verify sensor connectivity. If no valid response is detected, an error flag (`bmp_error = true`) is raised to trigger the automatic bus recovery routine.

Furthermore, in long-duration testing, I2C communication may occasionally be disrupted due to bus contention, electrical noise, or transient interference. To maintain stable data acquisition, the firmware implements an I2C bus recovery routine (`i2c_bus_recovery()`):

```

1  void i2c_bus_recovery() {
2      const int SDA_PIN = 20;
3      const int SCL_PIN = 21;
4
5      pinMode(SDA_PIN, INPUT_PULLUP);
6      pinMode(SCL_PIN, OUTPUT);
7
8      for (int i = 0; i < 9; i++) {
9          digitalWrite(SCL_PIN, LOW);
10         delayMicroseconds(5);
11         digitalWrite(SCL_PIN, HIGH);
12         delayMicroseconds(5);
13     }
14
15     Wire.end();
16     delay(100);
17     Wire.begin(); // Restart I2C communication
18     Serial.println("I2C_bus_recovery_executed_successfully.");
19 }

```

This mechanism temporarily reconfigures the SDA and SCL pins as general-purpose input/output (GPIO) to generate clock pulses and release the bus before reinitializing communication. This routine ensures that the sensor can automatically recover from transient faults without manual reset or power cycling.

4.3 Software Architecture and Control Logic

This section describes the software framework that governs the automated operation of the test platform. All components, including the peristaltic pump, UV LED module, BMP280 sensor, and touchscreen display, are coordinated through a modular control program running on the Arduino Giga R1 WiFi. The firmware is designed for deterministic task scheduling and synchronized communication between the hardware and the graphical interface. By employing event-driven logic, the system executes repeated urination cycles, real-time temperature monitoring, and device status updates without interrupting the LVGL-based GUI, ensuring stable and reproducible test operation.

Figure 12 provides an overview of the software architecture of the automated test bench in the form of a unified modeling language (UML) component diagram. The user interacts with the system via the touchscreen-based Test Bench GUI, which is implemented using the LVGL graphics library on the Arduino Giga Display Shield. Internally, the GUI Control component coordinates three functional tabs: the Setting tab configures the micturition cycles and cycle duration, the Device Status tab visualises the current state of the pump, UV LED, and environ-

mental sensors, and the Monitor tab displays elapsed time, completed cycles, and temperature trends. These GUI components use three core software modules: the Actuation Module (peristaltic pump and UV-LED control), the Environmental Monitoring module (BMP280 sensor), and the Error Recovery module (I2C bus recovery and sensor re-initialisation). All hardware-specific operations are encapsulated in the Arduino Hardware Driver, which accesses the Arduino Core API to control the Arduino Giga R1 WiFi, the Motor Shield, and the Display Shield and the connected pump, UV-LED ring, and BMP280 sensor. The following subsections describe these modules and their control logic in more detail.

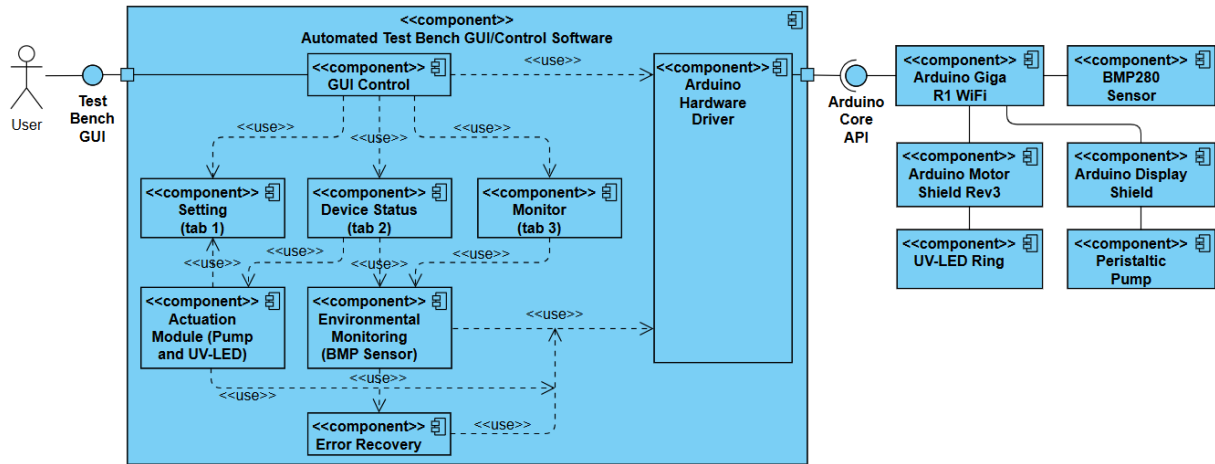


Figure 12: UML component diagram of the automated test bench software. The Test Bench GUI is managed by the GUI Control component, which connects to the Actuation, Environmental Monitoring, and Error Recovery modules. Hardware interaction is performed through the Arduino Hardware Driver and the Arduino Core API.

4.3.1 Program Structure and Modular Design

The control software of the automated test platform follows a modular architecture to ensure clarity, stability, and scalability. Each functional unit, including the peristaltic pump, UV LED module, BMP280 sensor, and touchscreen interface, is implemented as an independent code module. These modules communicate with the central control logic through clearly defined functions and shared parameters, allowing each subsystem to be tested individually while maintaining overall system consistency.

The software framework is divided into three main categories:

1. functional modules (.h and .cpp)
2. icon modules (.h and .c)
3. the main control program (main.ino)

Each category contributes to a specific level of functionality, ensuring that the program logic and hardware operation remain synchronized and maintainable.

4.3.1.1 Functional modules (.h / .cpp)

This group includes the pump, UV LED, temperature, and humidity modules, each responsible for a dedicated hardware function or data handling process. Every module defines its own control functions, which are called by the main program to perform specific actions. For example, `pump.h` and `pump.cpp` define `startMotor()` and `stopMotor()` for controlling the peristaltic pump; `uvled.h` and `uvled.cpp` contain `startLED()` and `stopLED()` to operate the sterilization light source; `temperature.h` and `temperature.cpp` acquire temperature data from the BMP280 sensor and trigger the recovery routine `i2c_bus.recovery()` in case of communication errors.

4.3.1.2 Icon modules (.h / .c)

The icon modules store graphical assets used in the LVGL-based user interface, such as icons representing the pump, UV LED, and temperature indicators. These resources are saved as `.h` and `.c` files and are loaded into memory during the initialization of the graphical interface. This approach reduces data access latency, ensures smoother rendering performance, and maintains a clean and portable code structure.

4.3.1.3 Main control program (main.ino)

The main program integrates all functional modules and manages the core system logic. During initialization, the program activates the display, touch interface, and sensors (`Display.begin()`, `TouchDetector.begin()`, `bmp.begin(0x76)`), and sets safe default states for all actuators (`stopMotor()`, `stopLED()`). Within the `loop()` function, the system executes non-blocking time control using the `millis()` function, coordinating the actuation cycles of the pump and UV LED while continuously updating temperature readings and user interface elements. The program also incorporates an event-driven callback system based on LVGL to handle user interactions such as button presses, slider adjustments, and confirmation dialogs. These callbacks ensure that all GUI actions are reflected immediately in the physical control logic, keeping the system responsive and consistent during long-term automated testing.

This modular and layered software structure achieves clear separation between functionality, visualization, and control coordination. The functional modules handle hardware operations, the icon modules manage visual assets, and the main program oversees timing and data exchange.

4.3.2 Main Control Loop and Task Scheduling

The main control loop is responsible for coordinating all system activities over time, including cycle automation, sensor data acquisition, and graphical interface updates. The program adopts a non-blocking architecture based on the `millis()` function, allowing multiple concurrent operations such as sensor reading, display refresh, and actuator control without delay.

4.3.2.1 Initialization Phase

In the `setup()` function, all modules are initialized and set to safe default states. The display and touch systems are started using `Display.begin()` and `TouchDetector.begin()`, while the BMP280

sensor is initialized with `bmp.begin(0x76)`. The pump and UV LED are both deactivated at startup to ensure electrical safety.

```

1 void setup() {
2   Serial.begin(115200);
3   Display.begin();
4   TouchDetector.begin();
5   Wire.begin();
6
7   // Initialize pump and LED modules
8   pinMode(motorIn1, OUTPUT);
9   pinMode(motorIn2, OUTPUT);
10  pinMode(motorENA, OUTPUT);
11  stopMotor();
12
13  pinMode(ledIn1, OUTPUT);
14  pinMode(ledIn2, OUTPUT);
15  pinMode(ledENA, OUTPUT);
16  stopLED();
17
18  if (!bmp.begin(0x76)) {
19    Serial.println("BMP280 not found");
20    bmp_error = true;
21  } else {
22    bmp_error = false;
23  }
24 }

```

This initialization ensures that all actuators remain inactive before operation begins and that any sensor communication failure is detected for later recovery.

4.3.2.2 Cycle Control

The system continuously executes its main logic in the `loop()` function using `millis()` as the timing reference. This method avoids delays and allows the system to perform independent tasks concurrently. The user sets two parameters on the touchscreen interface: the number of daily cycles and the duration of each cycle. The interval between two consecutive cycles is automatically calculated as

$$t_{\text{interval}} = \frac{86400}{N_{\text{cycles}}} \quad (4.1)$$

where N_{cycles} is the number of micturition cycles per day

```

1 int cycles_per_day = lv_slider_get_value(micturition_slider);
2 int cycle_duration = lv_slider_get_value(cycle_duration_slider);
3 int interval_between_cycles = 86400 / cycles_per_day;
4
5 if (!motor_is_running && (current_seconds - last_cycle_time >=
6   interval_between_cycles)) {
7   startMotor();
8   updatePumpStatus(true);
9   motor_is_running = true;
10  last_cycle_time = current_seconds;
11 }

```

```

11
12 if (motor_is_running && (current_seconds - last_cycle_time >=
13     cycle_duration)) {
14     stopMotor();
15     updatePumpStatus(false);
16     motor_is_running = false;
17     completed_cycles++;
18 }

```

This logic ensures that the pump starts and stops automatically based on the configured parameters. After each completed cycle, the system increments the count of completed cycles, providing continuous and repeatable operation.

4.3.2.3 Event-driven GUI Control

The user interface operates under an event-driven architecture based on the LVGL library. The main control button alternates between “START” and “STOP,” allowing the user to begin or end the simulation safely. When “STOP” is pressed, a confirmation dialog is displayed to prevent accidental interruption.

```

1 static void event_handler(lv_event_t * e) {
2     lv_event_code_t code = lv_event_get_code(e);
3     lv_obj_t * btn = lv_event_get_target(e);
4     lv_obj_t * label = lv_obj_get_child(btn, 0);
5     const char * text = lv_label_get_text(label);
6
7     if (code == LV_EVENT_CLICKED) {
8         if (strcmp(text, "STOP") == 0) {
9             show_confirm_dialog(btn);
10        } else {
11            lv_label_set_text(label, "STOP");
12            is_running = true;
13            startLED(); // Activate UV LED
14            start_time = millis();
15        }
16    }
17 }

```

This interactive structure provides an additional safety layer by confirming user commands and ensuring that both the pump and UV LED are shut down before the system enters an idle state.

4.3.2.4 Sensor Data Acquisition and Error Recovery

The system periodically reads temperature data from the BMP280 sensor. If repeated invalid readings or communication errors are detected, the `i2c_bus_recovery()` function is executed to reset the I2C bus and reinitialize communication.

```

1 if (bmp_error || isnan(bmp.readTemperature())) {
2     i2c_bus_recovery();
3     if (bmp.begin(0x76)) {
4         bmp_error = false;
5         Serial.println("BMP280 recovered successfully");
6     } else {

```

```
7     Serial.println("BMP280 still not responding");
8   }
9 }
```

This automatic recovery ensures long-term communication stability and prevents data loss during extended automated testing.

4.3.2.5 Status Updates and Safety Logic

At each loop iteration, the system updates the GUI to reflect the current operation status. If the system is not actively running, all actuators are forced into the OFF state, and the display immediately reflects the hardware state.

```
1  if (is_running) {
2    lv_label_set_text(elapsed_time_label, formatted_time);
3    lv_label_set_text(cycles_completed_label, String(completed_cycles)
4                      .c_str());
5  } else {
6    stopMotor();
7    stopLED();
8  }
```

This synchronization between software and hardware states prevents unintended motion or residual current after shutdown, ensuring safe transitions between operating modes.

Overall, the main control loop integrates timing management, user interaction, and fault-tolerant communication into a unified operational framework. By combining non-blocking control, event-driven architecture, and automatic recovery, the system achieves stable, long-term functionality while maintaining responsive performance.

4.4 Graphical User Interface Implementation

This section presents the design and implementation of the touchscreen GUI developed for the automated test platform. The GUI was implemented using the LVGL [24] in conjunction with the Arduino Giga Display Shield [29], providing an interactive environment for configuring test parameters, monitoring real-time data, and controlling device operation. The primary design goals were clarity, reliability, and intuitive interaction, ensuring that users can efficiently operate the system without prior programming knowledge.

4.4.1 Interface Layout and Navigation Structure

The user interface uses a three-tab layout that organizes system configuration, device monitoring, and data visualization. The Setting tab defines daily micturition cycles and cycle duration, the Device Status tab shows the current state of the pump, UV LED, and sensors, and the Monitor tab displays real-time temperature and cycle progress during testing..

Each tab was designed to fulfill a distinct operational purpose while maintaining consistency in color scheme, typography, and layout spacing. The tab was implemented as a separate LVGL

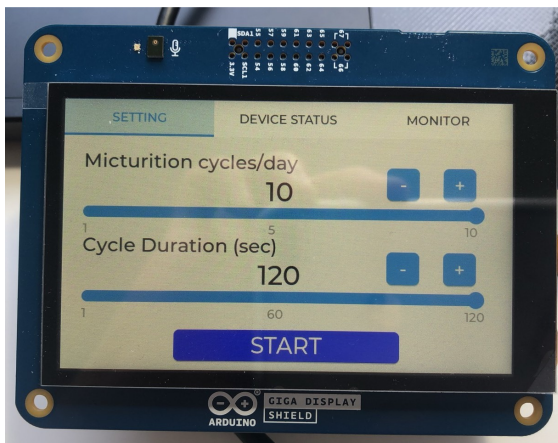
object hierarchy to simplify event handling and reduce UI latency. This approach ensures stable interaction even when background tasks—such as pump actuation or temperature acquisition—are running concurrently.

4.4.1.1 Setting Tab

The first tab, “Setting”, provides two adjustable parameters: the number of micturition cycles per day and the duration of each cycle. These parameters are controlled through LVGL sliders and numeric labels that update in real time. Users can fine-tune values using increment and decrement buttons.

```
1 lv_obj_t * tabview = lv_tabview_create(lv_screen_active());
2 lv_obj_t * tab1 = lv_tabview_add_tab(tabview, "SETTING");
3 create_named_slider(tab1, "Micturition_cycles/day", 1, 10, "", 20,
4     10, 1);
5 create_named_slider(tab1, "Cycle_Duration_(sec)", 1, 120, "", 20,
6     160, 1);
7 create_start_button(tab1, "START", &lv_font_montserrat_40, 0, 0);
```

Figure 13 shows the architecture of the setting tab. A central “START” button initiates the automated testing sequence (Figure 13 a)). When pressed, it triggers the `event_handler()` function, which activates the pump and UV LED, begins cycle timing, and changes the button label to “STOP”. If the user presses “STOP”, a confirmation window (`show_confirm_dialog()`) appears to prevent unintended interruption (Figure 13 b)).



a) Setting tab



b) Accidental touch protection confirm window

Figure 13: Touchscreen interface of the setting tab showing parameter configuration (left) and the confirmation window when stopping an active cycle (right)

4.4.1.2 Device Status Tab

The second tab, “Device Status”, summarizes the current operational state of all hardware components. Individual status cells are implemented using a custom helper function `create_status_cell()`, which combines an icon, label, and dynamic value text to represent each device’s condition.

```

1 create_status_cell(tab2, &uvled, "UV LED", "OFF", lv_palette_main(
  LV_PALETTE_RED), 20, 20);
2 create_status_cell(tab2, &pump, "Pump", "OFF", lv_palette_main(
  LV_PALETTE_RED), 400, 20);
3 create_status_cell(tab2, &temperature, "Temp", "NaN",
  lv_palette_main(LV_PALETTE_BLUE), 20, 200);

```

Figure 14 shows the architecture of the Device Status tab. When the pump or LED state changes, the corresponding LVGL label updates immediately, providing synchronized feedback between hardware actions and on-screen information. The displayed indicators (“ON” or “OFF”) are color-coded, with blue representing the active state and red representing the inactive state. Temperature readings from the BMP280 sensor are refreshed once per second and displayed in real time, while the humidity field is reserved for future integration of an additional sensor module.

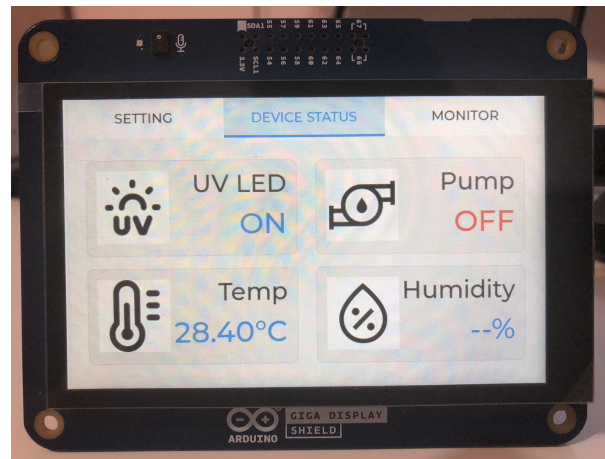


Figure 14: Touchscreen interface of the device status tab showing parameter of peripheral devices

4.4.1.3 Monitor Tab

The third tab, “Monitor”, visualizes environmental data and test progress. A line chart is generated using `lv_chart_create()` to show temperature trends collected from the BMP280 sensor. Each new data point is appended using `lv_chart_set_next_value()`, allowing continuous display updates during long-term operation. Real-time elapsed time and completed cycle count are displayed at the top of the screen, helping users verify that the system runs according to the defined schedule (Figure 15).

```

1 lv_obj_t * chart = lv_chart_create(tab3);
2 lv_chart_set_range(chart, LV_CHART_AXIS_PRIMARY_Y, 20, 40);
3 lv_chart_series_t * ser = lv_chart_add_series(chart, lv_palette_main(
  LV_PALETTE_BLUE), LV_CHART_AXIS_PRIMARY_Y);
4 lv_chart_set_next_value(chart, ser, temperature_value);

```

In summary, the developed automated test platform integrates actuation, sensing, and graphical control within a unified embedded architecture. By combining the Arduino Giga R1 WiFi,

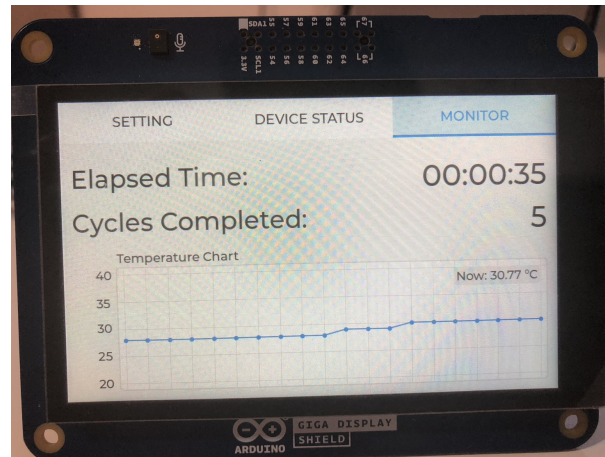


Figure 15: Touchscreen interface of the monitor tab showing the test progress and temperature trends

Motor Shield, and Display Shield, the system achieves stable and repeatable simulation of urination cycles with real-time monitoring and intuitive user interaction through an LVGL-based interface. The modular software design and fault-tolerant control logic ensure reliable long-term operation. This implementation provides a solid foundation for evaluating implantable devices under realistic physiological conditions. The following chapter (Chapter 5) presents the evaluation of the system's functionality and interface usability according to Jakob Nielsen's 10 Usability Heuristics, validating the practical performance of the proposed platform under real testing conditions.

5 Results

In this chapter, we present an expert-based heuristic evaluation grounded in Jakob Nielsen's ten usability principles [11], conducted without the involvement of external participants. The assessment systematically examined the interface's structural organization and terminological consistency, the visibility of system status and latency of feedback, the clarity and correctness of operational workflows, and the empirical adequacy of on-screen presentation and control effectiveness. All findings are reported as objective observations; interpretive analysis and design implications are reserved for the subsequent Discussion.

5.1 Heuristic Evaluation Results According to Jakob Nielsen's 10 Usability Heuristics

For each heuristic, we first articulate a concise definition derived from Jakob Nielsen's "10 Usability Heuristics for User Interface Design" (Nielsen Norman Group, April 24, 1994) [11]. The evaluation is conducted based on the GUI implementation results presented in Chapter 4 (Figures 13-15), systematically examining how the interface structure, terminology consistency, visibility of system status, clarity of operation flow, and visual presentation align with the defined usability principles. Each assessment focuses on observable interface behaviors and visual responses, rather than user-reported feedback, to objectively determine the extent of conformity to each principle.

5.1.1 Visibility of System Status

A well-designed keeps users continuously informed about system state through timely, appropriate feedback, so they can understand the outcome of their actions, decide next steps with confidence, and develop predictable, trust-building interactions with the product and brand.

The graphical interface delivers clear and immediate visual feedback that allows users to continuously perceive the system's operational status, effectively demonstrating the principle of Visibility of System Status. In the SETTING tab, when the simulation is initiated, the main control button label changes instantly from Start to Stop and is accompanied by a brief touch animation, confirming that the command has been successfully triggered. This immediate response ensures that user interactions are acknowledged without delay, reinforcing the sense of system responsiveness and control.

In the DEVICE STATUS and MONITOR tabs, the system continuously communicates its ongoing state through dynamic visual indicators. In the DEVICE STATUS tab, the pump and UV-LED indicators update in real time to reflect their actual hardware conditions, using synchronized color and text changes to clearly display their ON/OFF states. In the MONITOR tab, the elapsed time, cycle count, and temperature curve are displayed and refreshed continuously, allowing users to intuitively observe both environmental variations and overall testing progress. Together, these visual elements create a consistent and reliable feedback loop that keeps users informed of system activity at all times.

5.1.2 Match Between the System and the Real World

The system should communicate using the user's own language, concepts, and conventions rather than internal technical terms. Interfaces that follow real-world mappings and logical sequences make actions more intuitive, helping users understand, learn, and remember how to operate the system naturally

The graphical interface of the test platform is designed based on laboratory operation logic rather than internal technical terminology, ensuring that the system communicates using language and concepts familiar to users. All labels, units, and on-screen prompts adopt expressions commonly used in real experimental environments. For instance, parameter settings are expressed with intuitive physical quantities such as "cycles per day" and "seconds per cycle" instead of abstract system variables. Temperature values are displayed in degrees Celsius, consistent with standard laboratory measurement units. The peripheral components are represented by clear, recognizable icons, allowing users to immediately identify each component.

The overall workflow follows the logical sequence of an actual laboratory procedure. Users first set the parameters, then start the test, observe environmental conditions in real time during operation. This cause-and-effect order aligns with users' mental models of fluid simulation and device validation, making the operation intuitive and easy to remember. From a usability perspective, the close correspondence between interface language and real operational context effectively reduces the learning curve and prevents cognitive gaps between "system language" and "user language." The overall design ensures that system behavior matches user expectations, resulting in a natural and fluent interaction experience.

5.1.3 User Control and Freedom

Users should always feel in control of the interface and be able to easily undo or exit unintended actions. Providing clear escape options and reversible processes promotes confidence, reduces frustration, and prevents users from feeling trapped in unwanted workflows.

The graphical user interface demonstrates a high level of safety and consistency in relation to the principle of User Control and Freedom. The overall workflow is clearly structured, allowing users to remain aware of the system's status at all times and to safely interrupt or reverse any action without compromising operational stability. During simulation, the main control button is prominently labeled Stop, clearly indicating that the process can be terminated at any moment. When pressed, the system does not stop immediately but instead displays a confirmation window (Figure 13 b)), prompting the user to verify the intention to end the operation. This dual-confirmation mechanism effectively prevents accidental interruptions and ensures that termination remains a deliberate and reversible process.

After confirmation, the interface promptly returns to the idle state. The visual indicators for the pump and UV-LED switch to their inactive colors, and the monitoring curve stops updating in real time. These immediate visual transitions clearly communicate that the system has safely shut down and reset to a ready-to-operate condition. From an observer's perspective, the interface combines clear control entry points, instant visual feedback, and a secure confirmation mechanism, ensuring that every action is visible, predictable, and reversible. This design allows users to maintain full control and confidence throughout the entire operational process, fully embodying Nielsen's User Control and Freedom heuristic.

5.1.4 Consistency and Standards

Consistent terminology, behavior, and layout across the interface help users understand patterns and reduce cognitive effort. Following established platform and industry conventions aligns with user expectations shaped by other systems, making the design more predictable and easier to learn.

The overall graphical interface demonstrates a high level of consistency in both visual presentation and interaction behavior. Text elements are clearly categorized according to their purpose and semantic function: Setting, Device Status, and Monitor belong to the navigation category; Start and Stop represent control commands; while Elapsed Time and Cycles Completed serve as status indicators. Each category follows a unified design language in terms of color scheme, font style, and font size, ensuring a clear visual hierarchy and well-defined information emphasis. Interactive elements such as buttons, sliders, and labels maintain consistent positions and appearances across all tabs. Their behaviors are predictable and uniform, allowing users to quickly identify functional areas without the need to readjust or relearn the interface, thereby reducing cognitive effort and learning time.

The color and labeling logic of the status indicators are equally consistent: during operation, blue with the label ON signifies an active state, while red with OFF indicates a stopped state. This fixed and intuitive mapping allows users to recognize system conditions instantly and accurately. Additionally, the overall layout follows conventions commonly used in embedded control panels and industrial monitoring systems, establishing a clear hierarchy and logical grouping of information. From an observer's perspective, this consistency in terminology, layout, and interaction logic effectively minimizes cognitive load and enhances operational fluency and predictability.

5.1.5 Error Prevention

Preventing errors is more effective than simply handling them after they occur. The design should minimize error-prone situations through validation, confirmation steps, and logical constraints that eliminate slips and mistakes before they can happen.

The graphical interface clearly embodies the principle of Error Prevention by proactively minimizing user mistakes through visual constraints, confirmation mechanisms, and explicit instructional text. When the system is actively running a test and the user attempts to stop the process, a confirmation window automatically appears with clear textual prompts asking for explicit confirmation before the action is executed. The stop command takes effect only after the user deliberately presses "YES." This mechanism prevents accidental interruptions during ongoing testing and ensures that all critical operations are intentional and controlled by the user.

During the configuration phase, the sliders for setting the number of cycles and the duration of each test are limited to predefined valid ranges, preventing invalid or excessive input. Real-time numerical feedback and descriptive labels are displayed next to the sliders, helping users clearly understand and verify their chosen parameters. The interface also maintains consistent color schemes, fonts, and labeling across all control elements, each accompanied by concise textual hints. Buttons and sliders are intuitively named according to their functions, reducing ambiguity and guiding users toward correct operation. These clear textual cues, combined with

logical interaction constraints, make the workflow predictable and prevent errors before they occur.

5.1.6 Recognition Rather than Recall

Interfaces should reduce the user's memory load by keeping essential options, labels, and actions visible or easily accessible. Designs that promote recognition rather than recall allow users to focus on the task instead of memorizing information across different screens.

The graphical interface maintains the visibility of key information and operational options, effectively reducing the user's memory load and making the operation process more intuitive without relying on recall. Core functions such as pump control, UV-LED activation, and temperature monitoring are presented through clear icons and text labels that remain continuously visible on the screen. The Device Status tab combines color and text to display the real-time operational state of each component, allowing users to instantly recognize whether a device is active without having to recall previous settings or actions.

In the monitoring view, the temperature curve, elapsed time, and completed cycle count are displayed simultaneously within a single screen, while the current temperature value is also shown at the upper right corner of the temperature chart (Figure 15), further enhancing the immediacy and readability of the information. Users can therefore grasp the overall system status without switching between screens. Labels and icons across all tabs are designed consistently, with fixed positions and unified naming, enabling users to quickly relocate familiar functions after first recognition and maintain a smooth and efficient interaction experience.

5.1.7 Flexibility and Efficiency of Use

Good design accommodates both novice and expert users by providing efficient shortcuts, flexible interaction paths, and customizable workflows. This adaptability enables users to choose the most convenient method and maintain high performance as they gain experience.

The overall graphical interface is designed with clarity and simplicity at its core, allowing users of different experience levels to operate the system effectively within the same environment. All primary functions such as setting test parameters, starting and stopping operations, and monitoring real-time results are clearly arranged within a well-structured tab layout, enabling users to understand how to interact with the system without prior training or documentation.

For first-time users, the interface structure and on-screen cues guide them through configuration and execution in a very short time. Experienced users, on the other hand, can complete the entire workflow even more efficiently thanks to the familiar layout and consistent feedback mechanisms. Because the interface involves few steps, a clear operational sequence, and a distinct separation between control elements and status displays, the overall experience maintains a low learning curve and high operational efficiency. This design allows the system to be easily understood by beginners while supporting fast and streamlined performance for expert users.

5.1.8 Aesthetic and Minimalist Design

An effective interface focuses on clarity and simplicity, displaying only information that is directly relevant to the task. Every visual element should serve a purpose, ensuring that users can concentrate on key actions without distraction or cognitive overload.

The interface design of the system follows the principle of clarity and task-oriented minimalism, displaying only information that is directly relevant to operation and monitoring. It is organized into three tabs: Setting, Device Status, and Monitor, each maintaining a clean and intuitive structure without unnecessary text or decorative graphics. During configuration or observation, only essential controls and status indicators are presented, ensuring that the workspace remains clear and free from distraction.

Throughout operation, this minimalist logic is consistently applied. The Device Status tab uses only color and concise labels to indicate the states of the pump and UV-LED, while the Monitor tab presents a single graph showing temperature variation and cycle progress without redundant numerical or visual elements. By avoiding visual noise and ensuring that every component serves a specific function, the interface allows users to focus on key information and actions.

5.1.9 Help Users Recognize, Diagnose, and Recover from Errors

When errors occur, the system should communicate them in clear, human-readable language that identifies the issue and offers actionable solutions. Proper visual emphasis ensures errors are noticeable, allowing users to recover quickly and continue their tasks without confusion.

When the sensor communication is temporarily interrupted, the temperature field on the display immediately shows “NaN,” clearly indicating that the system is no longer receiving valid data from the sensor. This direct visual feedback helps the user instantly recognize that a connection error has occurred without requiring any technical knowledge or additional diagnostic messages. The “NaN” display serves as an intuitive signal that isolates the source of the problem, ensuring that the interface remains transparent about the current system state.

In response to this condition, the system automatically initiates an internal reconnection procedure to restore I2C communication. From an observer’s perspective, the temperature graph pauses momentarily while the reconnection takes place, confirming that the recovery process is active. Within a short time, the temperature values reappear and the graph resumes updating without any user intervention. This seamless transition from error to recovery demonstrates the interface’s ability to recognize, diagnose, and autonomously correct faults.

5.1.10 Help and Documentation

Although the ideal design minimizes the need for help, accessible and task-focused documentation remains essential. Such guidance should be concise, searchable, and provide step-by-step instructions that directly support users in completing their goals.

The graphical interface of this system follows a low-dependence-on-help design approach, meaning that it does not rely on traditional help documents or assistance modules to guide users. Instead, it enables intuitive task completion through clear structural organization and semantically meaningful interface elements. The entire layout is organized according to the

operational logic of the task: configuring parameters (Setting), monitoring device status (Device Status), and observing test results (Monitor). The naming and ordering of these tabs naturally correspond to the real operational sequence. Interface elements such as sliders, buttons, and labels use concise, task-oriented wording, allowing users to understand the purpose and flow of actions directly from the interface hierarchy and visual layout, without the need for additional explanations.

However, this design, while enhancing operational fluency and visual clarity, also means that the system does not include a formal Help and Documentation module. There is no dedicated “Help” button, tooltip dialog, or step-by-step tutorial integrated into the interface. Consequently, first-time users who are unfamiliar with the testing process or parameter meanings may need to rely on experimentation or external references to understand the workflow. In other words, the system’s usability is achieved through structural and linguistic clarity rather than supplemental documentation, reflecting a structure-guided design that emphasizes intuitive understanding over explicit instructional support.

In summary, the heuristic evaluation shows that the developed graphical interface largely conforms to Jakob Nielsen’s 10 usability principles. The system provides clear feedback, consistent visual design, and intuitive workflow alignment with real laboratory procedures, ensuring smooth and reliable operation. Its structure-guided design allows users to interact effectively without prior training, though it omits a formal Help and Documentation module. Overall, the interface demonstrates strong usability, clarity, and stability. The following chapter, Discussion (Chapter 6), further interprets these results and explores potential improvements.

6 Discussion

The prototype developed in this study successfully integrates actuation, sensing, and graphical control within a unified embedded environment. Compared with conventional laboratory simulation systems described in the literature, the proposed platform demonstrates excellent reproducibility, timing stability, and long-term operational consistency. As of this writing, the system has been running continuously for nearly two months and has successfully completed over 400 automated test cycles without any critical software errors or control failures. These results confirm that the system architecture and control logic can maintain stable operation over extended periods, demonstrating both its reliability and engineering maturity in simulating urination cycles and monitoring environmental conditions.

Compared with existing biofilm or urinary flow simulation systems, the developed prototype achieves a comparable level of functionality in cycle scheduling, environmental sensing, and data visualization, while offering high customizability and expandability. In dynamic bladder and catheter models described by Xiong et al. (2021) [19], laboratory setups often rely on large-scale circulation loops or external pressure controllers, limiting accessibility and reproducibility in smaller research settings. The presented platform offers a compact and open-source alternative that achieves many of the same functional objectives while remaining significantly easier to deploy and maintain. Similarly, Zheng et al. (2022) [27] emphasise that physiological bladder dynamics involve complex interactions between flow, pressure, and reflux; while the present system does not replicate these fully, it provides a robust and modular baseline for future enhancement. In terms of biofilm research methodology, Ramachandra et al. (2023) [25] and Yang et al. (2024) [26] highlight the growing need for dynamic, multi-parameter monitoring tools. The extensible design of the current platform positions it well for integration with additional sensing and actuation modules required for advanced biofilm studies.

From a usability perspective, the three-tab layout and consistent interface logic provide a user experience similar to that of professional monitoring systems, making the platform suitable for both research and routine laboratory operation. The graphical interface is particularly effective in conveying system status and guiding operation. Immediate visual feedback, clearly separated tab structures, and the continuous display of key parameters collectively create a highly transparent operational experience. Even without prior training, users can intuitively complete the entire process from parameter configuration to execution and monitoring based solely on the visual organization of the interface.

Despite these strengths, several limitations remain. First, the current platform does not support pressure-regulated or closed-loop flow control. Existing dynamic bladder models emphasise that shear stress, pressure oscillations, and retention time strongly influence microbial adhesion and detachment kinetics, and these parameters cannot yet be reproduced in the present system [19, 25]. Furthermore, the unidirectional pump configuration prevents simulation of reverse flow or reflux phenomena, which have been shown by Zheng et al. (2022) [27] to alter encrustation patterns and biofilm behaviour. Environmental monitoring is currently limited to temperature, whereas biofilm growth is also influenced by pH, turbidity, and ionic concentration [26]. Expanding the sensor suite would therefore significantly improve biological relevance.

Regarding system robustness, the BMP280 temperature sensor, while stable, provides only moderate resolution and may exhibit minor fluctuations during prolonged operation, slightly affecting long-term measurement precision. In rare cases, short communication delays or in-

terruptions can occur on the I2C bus. Although the built-in recovery mechanism restores functionality automatically, this still indicates potential for firmware-level optimization. Moreover, the interface currently lacks user activity logging, automated error reporting, and a formal help module. While the interface is intuitive, integrating lightweight onboarding guidance or usage analytics would support long-term usability evaluation based on Nielsen's heuristics.

Overall, the system has achieved its intended goals in reliability, clarity, and usability, demonstrating sustained stability and consistent performance throughout extended testing. Through modular design and a clear interface structure, this study has realized an automated testing platform with strong potential for laboratory application. Addressing the identified limitations, such as adding pressure regulation, multi-parameter sensing, and bidirectional flow, would further improve physiological realism and experimental capability. Corresponding development strategies and future extensions will be elaborated in Chapter 7, Summary and Future Prospects.

7 Summary and future prospects

This work presented the development of an automated test platform designed to support long-term investigations of biofilm formation on implantable urological devices. The system integrates mechanical actuation, environmental monitoring, and an embedded graphical interface within a cohesive hardware and software framework. Based on the Arduino Giga R1 WiFi, the platform combines a peristaltic pump, a UV-LED module, and environmental sensors, all controlled through an LVGL-based touchscreen interface that enables real-time monitoring and intuitive interaction. The modular software structure clearly separates GUI management, actuator control, sensor processing, and error recovery routines, providing reliable operation during extended testing and offering a transparent foundation for future experimental extensions.

Beyond its technical realization, this platform contributes to ongoing discussions in the literature concerning the need for accessible, reproducible, and customizable dynamic testing environments. Prior studies highlight that biofilm formation is strongly influenced by environmental fluctuations, shear forces, and retention dynamics, as described by Ramachandra et al. (2023) and Yang et al. (2024) [25, 26]. The platform developed in this work addresses these research needs by providing a structured baseline system that can be expanded towards more physiologically realistic conditions. Although the current setup focuses on unidirectional flow and temperature monitoring, its modularity offers a clear pathway for future inclusion of pressure regulation, cyclic loading profiles, and multiparameter sensing. These features are considered essential in dynamic bladder models reported by Xiong et al. (2021) and Zheng et al. (2022) [19, 27].

Future work will concentrate on enhancing both the functional depth and biological relevance of the platform. The network capabilities of the Arduino Giga R1 WiFi make it possible to implement remote supervision, automated data logging, and cloud-based analysis pipelines. Integrating additional sensors for parameters such as flow rate, pressure, pH, turbidity, or conductivity would allow the system to capture a wider range of environmental conditions that shape biofilm behaviour. On the control side, the introduction of adaptive cycle regulation or closed-loop feedback would support the reproduction of more complex physiological dynamics. With respect to the user interface, features such as instructional overlays, activity metrics, or error-reporting tools would enable more systematic usability evaluations and align the system more closely with established principles in medical-device interface design, as described by Nielsen (1994) and Zhang et al. (2003) [36, 33].

In the long term, the platform has the potential to evolve into a versatile experimental tool that extends beyond the testing of urological implants. By incorporating additional sensing and control modules, it may serve as a general framework for biomaterial evaluation, dynamic biofilm studies, or controlled fluid simulation in broader biomedical contexts. Its open-source architecture makes it accessible for academic laboratories seeking cost-efficient solutions, while its deterministic timing and modular design support integration with data-driven analysis methods and machine-learning-based predictive models. These developments would enable the system to serve as a foundation for more advanced forms of automated experimentation and biomedical-device assessment in future research.

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